



Data Communications and Networking

Fourth Edition

Chapter 14

Wireless LANs



14-1 IEEE 802.11

IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers.

Topics discussed in this section:

Architecture

MAC Sublayer

Physical Layer



IEEE 802 Standards Working Groups

Working Group	Task	Status
802.1	Higher Layer LAN Protocols Working Group	Active
802.2	Logical Link Control Working Group	Inactive
802.3	Ethernet Working Group	Active
802.4	Token Bus Working Group	Disbanded
802.5	Token Ring Working Group	Inactive
802.6	Metropolitan Area Network Working Group	Disbanded
802.7	Broadband TAG	Disbanded
802.8	Fiber Optic TAG	Disbanded
802.9	Isochronous LAN Working Group	Disbanded
802.10	Security Working Group	Disbanded
802.11	Wireless LAN Working Group	Active
802.12	Demand Priority Working Group	Inactive
802.14	Cable Modem Working Group	Disbanded
802.15	Wireless Personal Area Network (WPAN) Working Group	Active
802.16	Broadband Wireless Access Working Group	Active
802.17	Resilient Packet Ring Working Group	Active
802.18	Radio Regulatory TAG	Active
802.19	Coexistence TAG	Active
802.20	Mobile Broadband Wireless Access (MBWA) Working Group	Active
802.21	Media Independent Handoff Working Group	Active
802.22	Wireless Regional Area Networks	Active

IEEE 802.16e :

(Mobile) Broadband Wireless Access

WiMAX, an acronym that stands for Worldwide Interoperability for Microwave Access(802.16)

WiFi (Wireless Fidelity) : 802.11 TAG (Technical Advisory Group)

15.1 IEEE 802.11 - Architecture

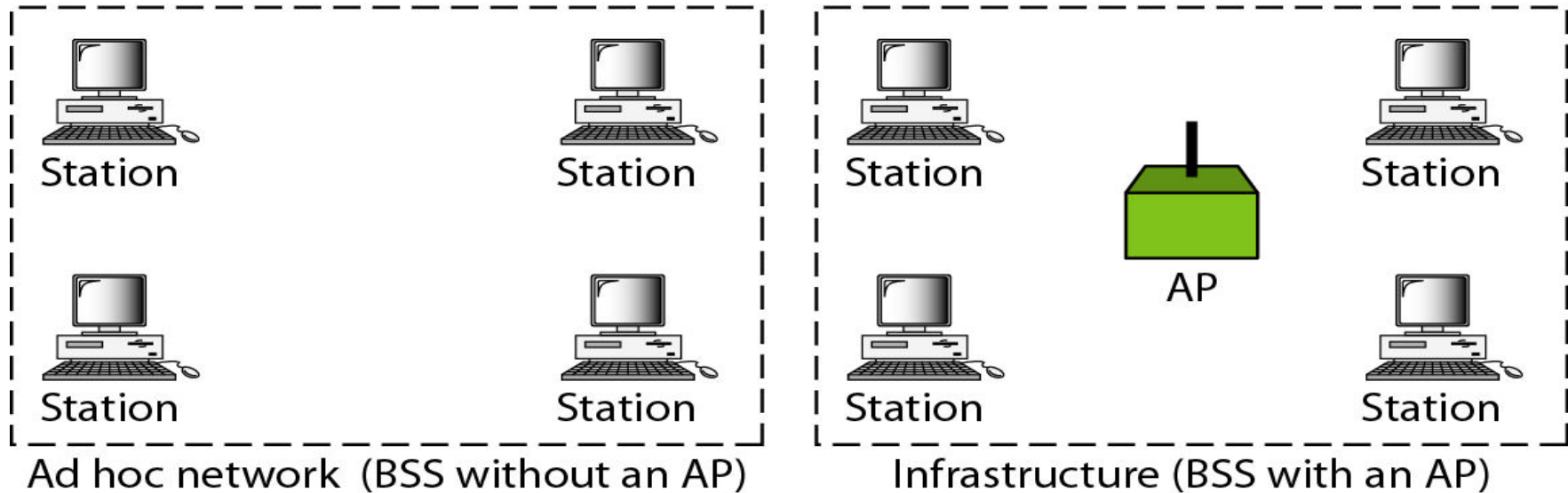
- ❑ Basic Service Set (BSS)
- ❑ Extended Service Set (ESS)



Basic Service Set

BSS: Basic service set

AP: Access point



- made up of stationary or mobile wireless stations, and possibly a central base station (access point)
- BSS without an AP is stand-alone, cannot send data to other BSSs

Basic Service Set

Note

A BSS without an AP is called an **ad hoc** network;
a BSS with an AP is called an **infrastructure** network.

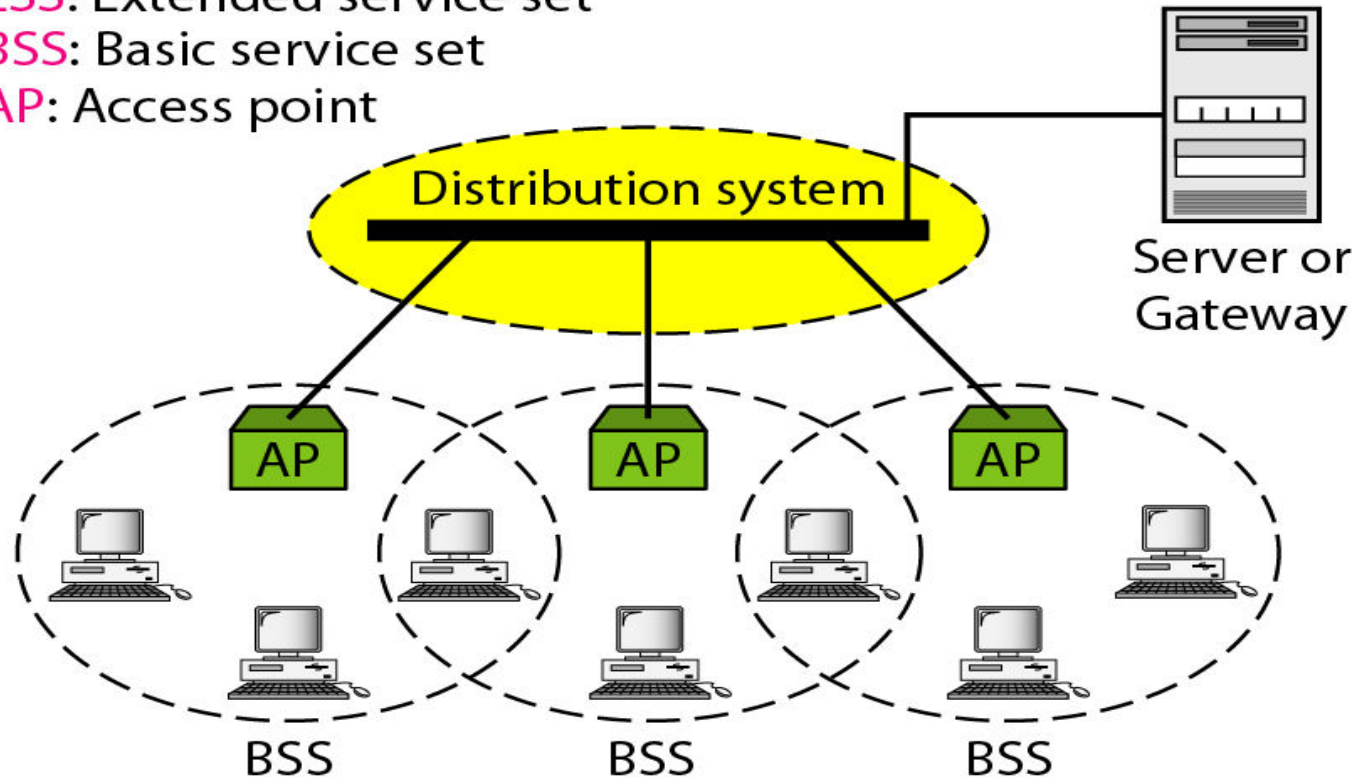


ESS – Extended Service Set

ESS: Extended service set

BSS: Basic service set

AP: Access point



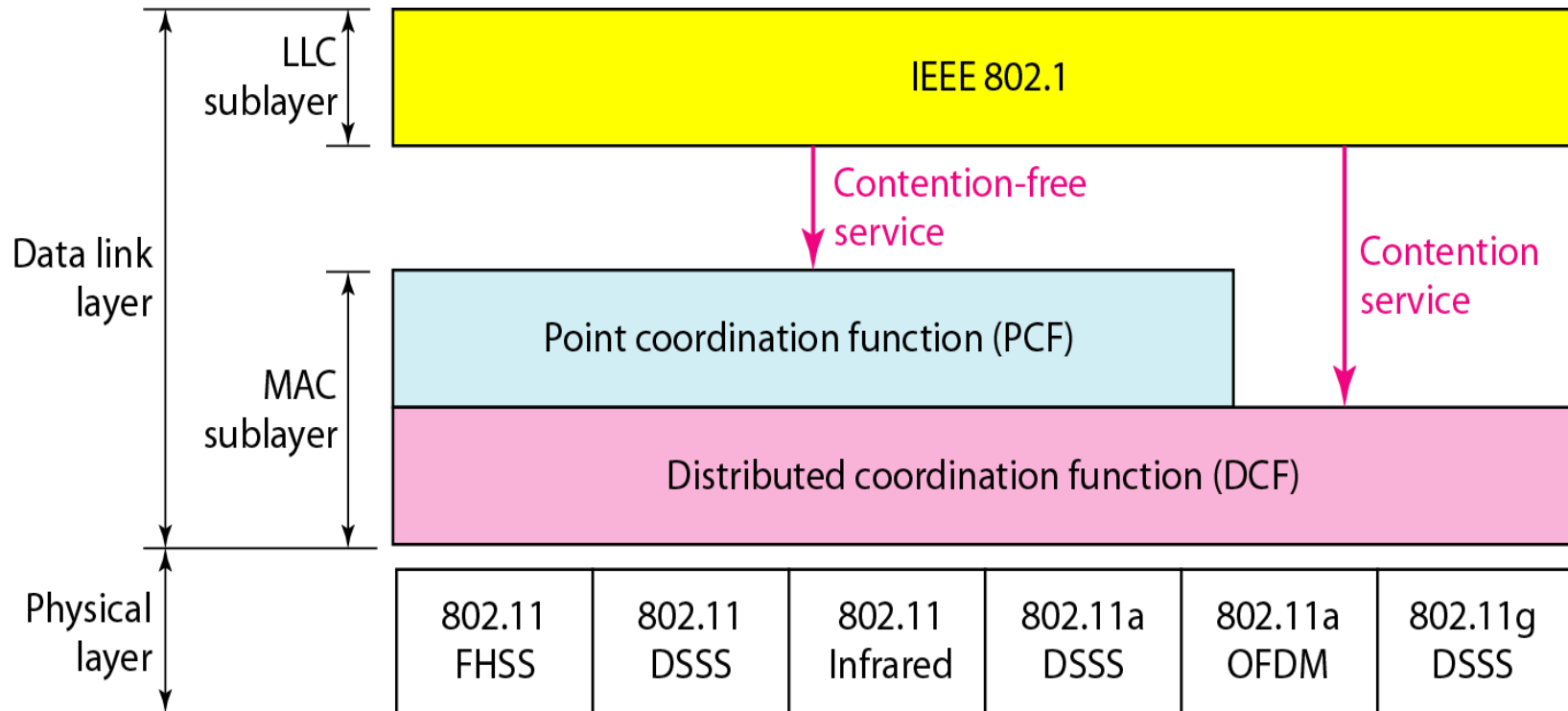
- ❑ Made up of two or more BSSs with APs – connected through a distribution system (usually a wired LAN)

Station Types

- ❑ **No-transition mobility – either stationary or moving only inside a BSS**
- ❑ **BSS-transition mobility – can move from one BSS to another, confined to one ESS**
- ❑ **ESS-transition mobility – can move from one ESS to another**

MAC Sublayer

Figure 14.3 *MAC layers in IEEE 802.11 standard*



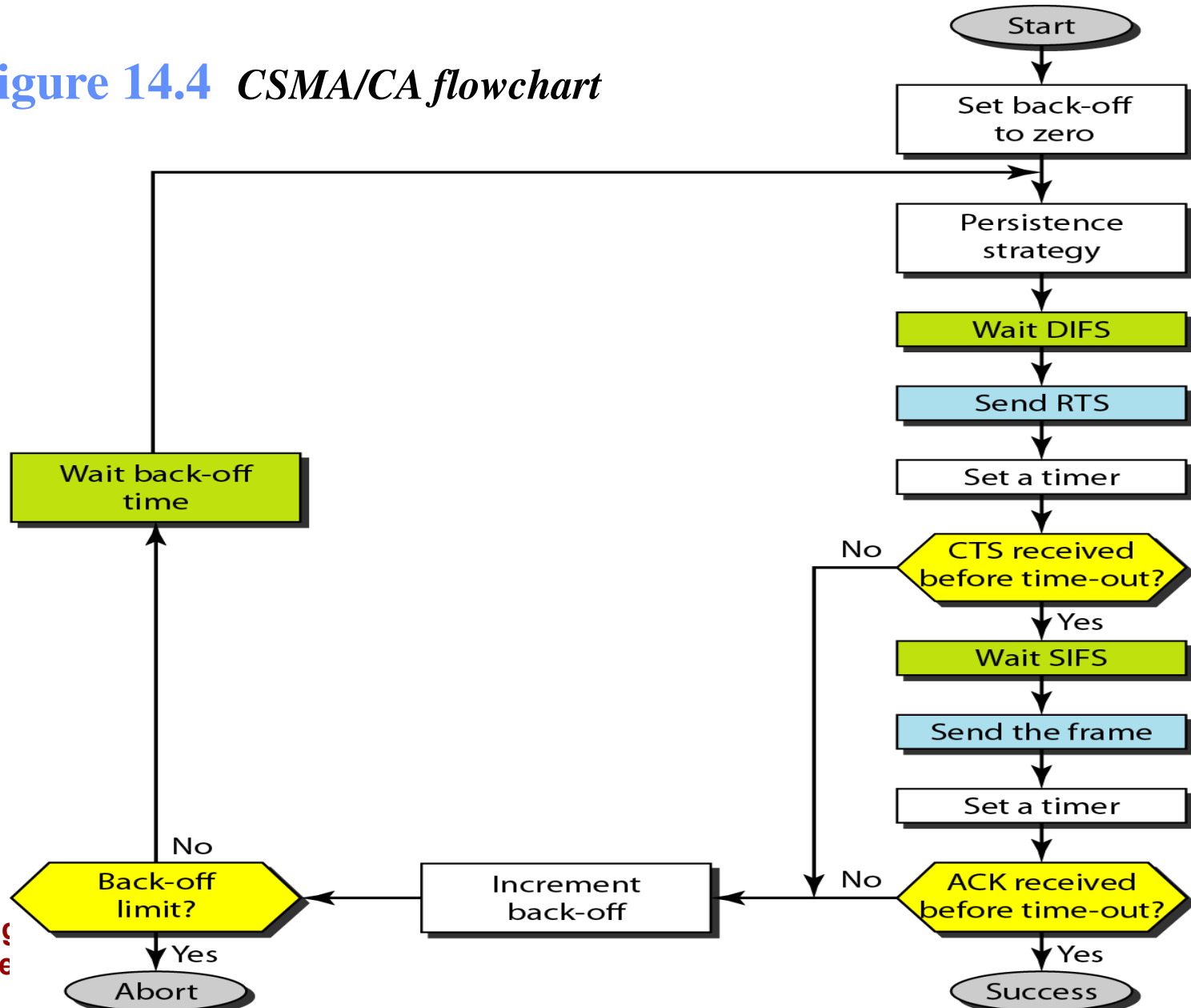
- ❑ **802.11 defines the specs for the conversion of bits to a signal**

Distribution Coordination Function

- ❑ One of the two protocols defined by IEEE at the MAC sublayer is called the Distributed Coordination Function (DCF)
 - ❖ DCF uses CSMA/CA as the access method.
 - ❖ Wireless LANs cannot implement CSMA/CD for three reasons:
 1. For collision detection a station must be able to send data and receive collision signals at the same time. This can mean costly stations and increased bandwidth requirements.
 2. Collision may not be detected because of the hidden station problem.
 3. The distance between stations can be great. Signal fading could prevent a station at one end from hearing a collision at the other end.

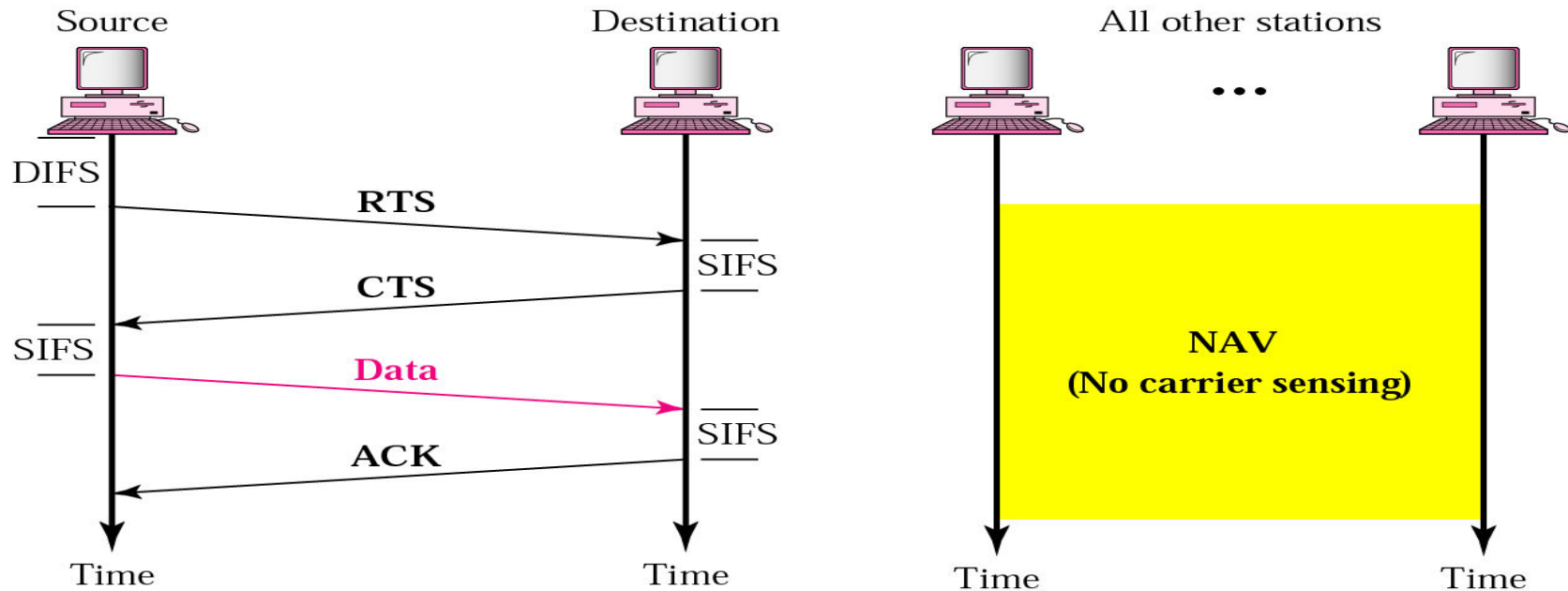
CSMA/CA Process flowchart

Figure 14.4 CSMA/CA flowchart



Frame Exchange Time Line

Figure 14.5 *CSMA/CA and NAV*



DIFS – distributed interframe space, NAV – network allocation vector

SIFS – short interframe space, RTS – request to send, CTS – clear to send

*** Collision only occurs during handshake period (RTS, CTS)**

PCF

Point Coordination Function (PCF)

- ❖ **PCF is an optional access method that can be implemented in an infrastructure network.**
- ❖ **It is implemented on top of the DCF and is used mostly for time-sensitive transmission.**

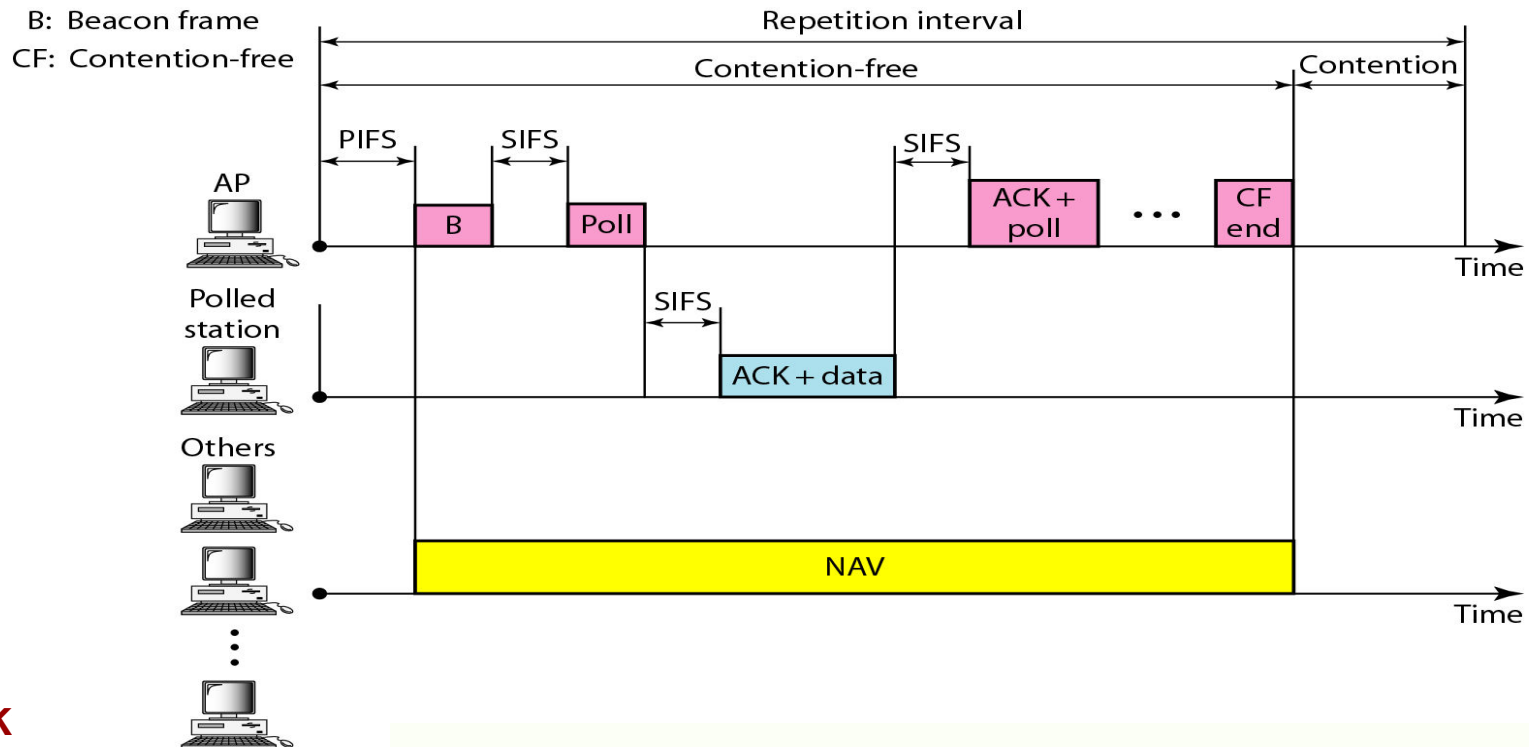
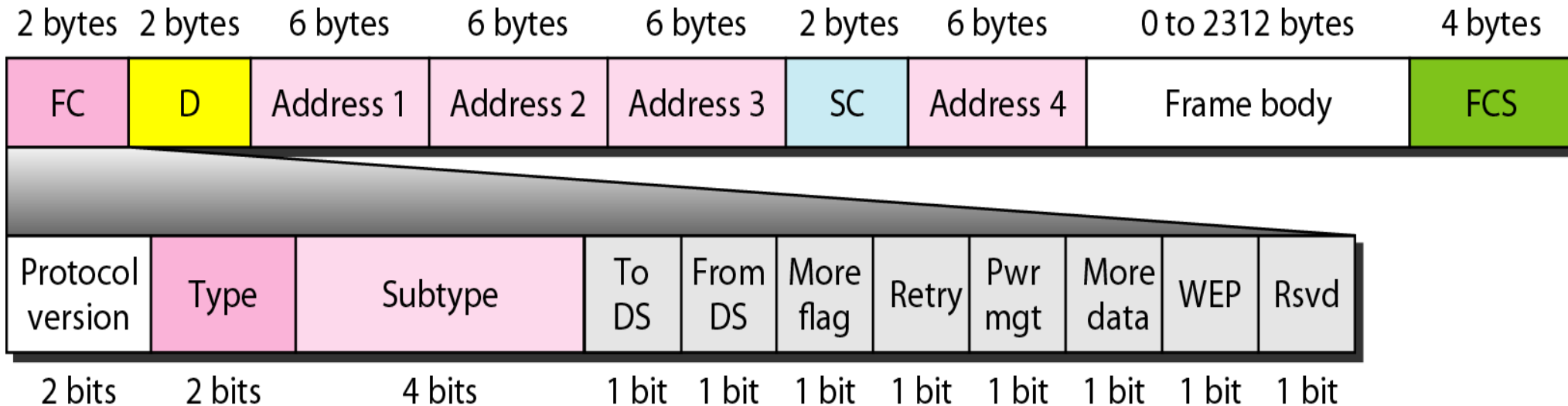


Figure 14.6 *Example of repetition interval*

Frame format



FC : frame control

D: the duration of the transmission that is used to set the value of NAV

SC: Sequence Control – defining the sequence number of the frame to be used in flow flow control



Frame format

Table 14.1 *Subfields in FC field*

<i>Field</i>	<i>Explanation</i>
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 14.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

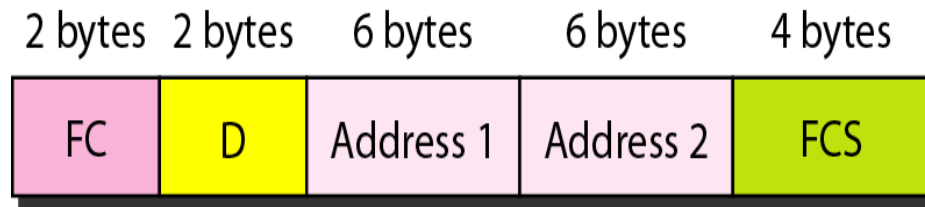


Management Frames

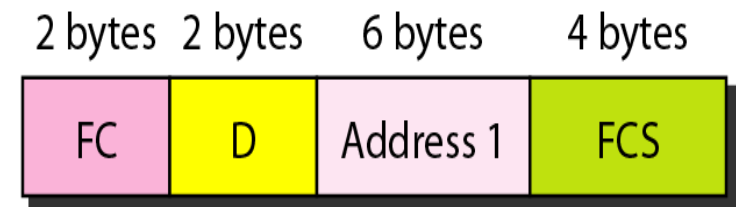
- ❑ A wireless LAN in IEEE 802.11 has 3 categories
 - ❖ Management Frame : 00
 - Used for the initial communication between stations and access points.
 - ❖ Control Frame : 01
 - Used for accessing the channel and acknowledging frames.
 - ❖ Data Frame : 10
 - Used for carrying data and control information.



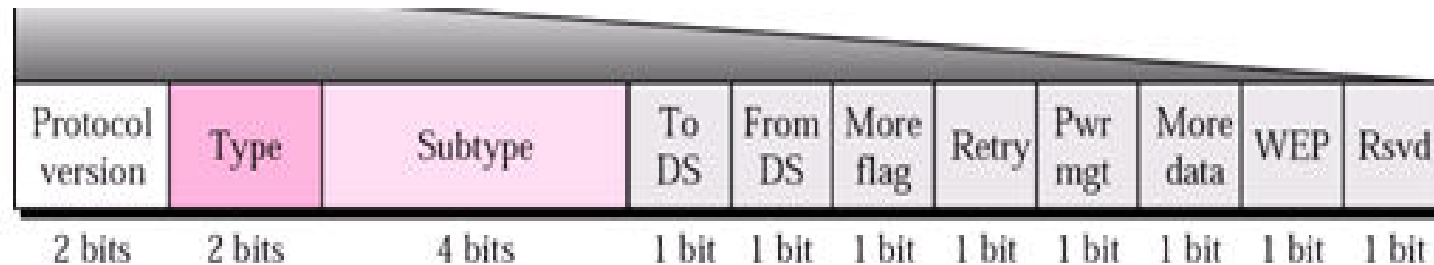
Control frames



RTS



CTS or ACK



Subtype values of subfields in control frames

Table 14.2 *Values of subfields in control frames*

□ **type field : 01**

<i>Subtype</i>	<i>Meaning</i>
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)



Addressing Mechanism

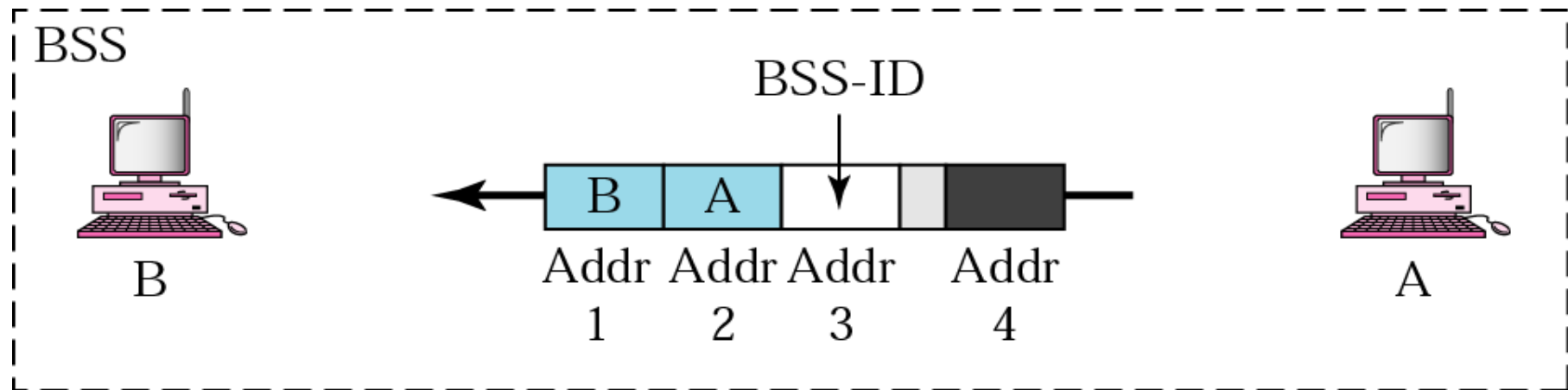


Table 14.3 *Addresses*

<i>To DS</i>	<i>From DS</i>	<i>Address 1</i>	<i>Address 2</i>	<i>Address 3</i>	<i>Address 4</i>
0	0	Destination	Source	BSS ID	N/A
0	1	Destination	Sending AP	Source	N/A
1	0	Receiving AP	Source	Destination	N/A
1	1	Receiving AP	Sending AP	Destination	Source

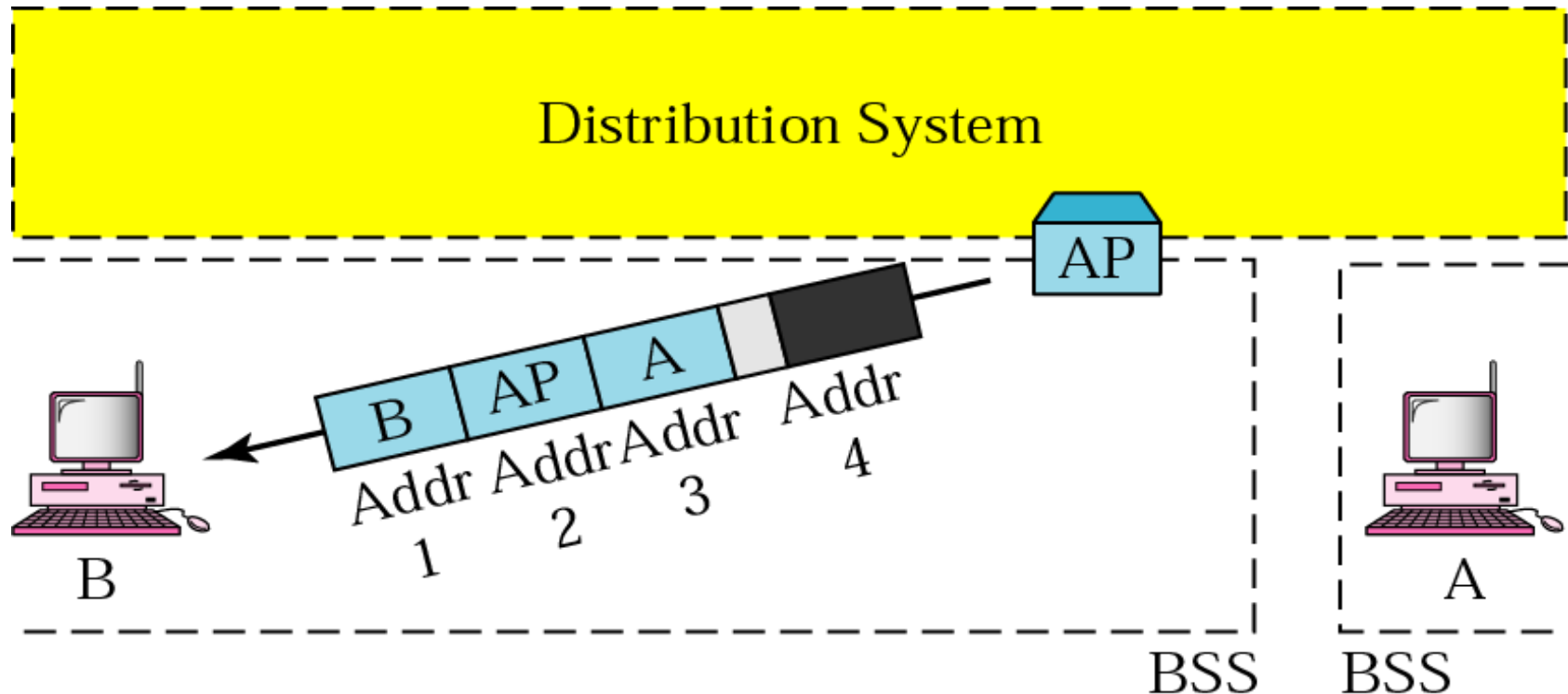


Addressing mechanism: case 1



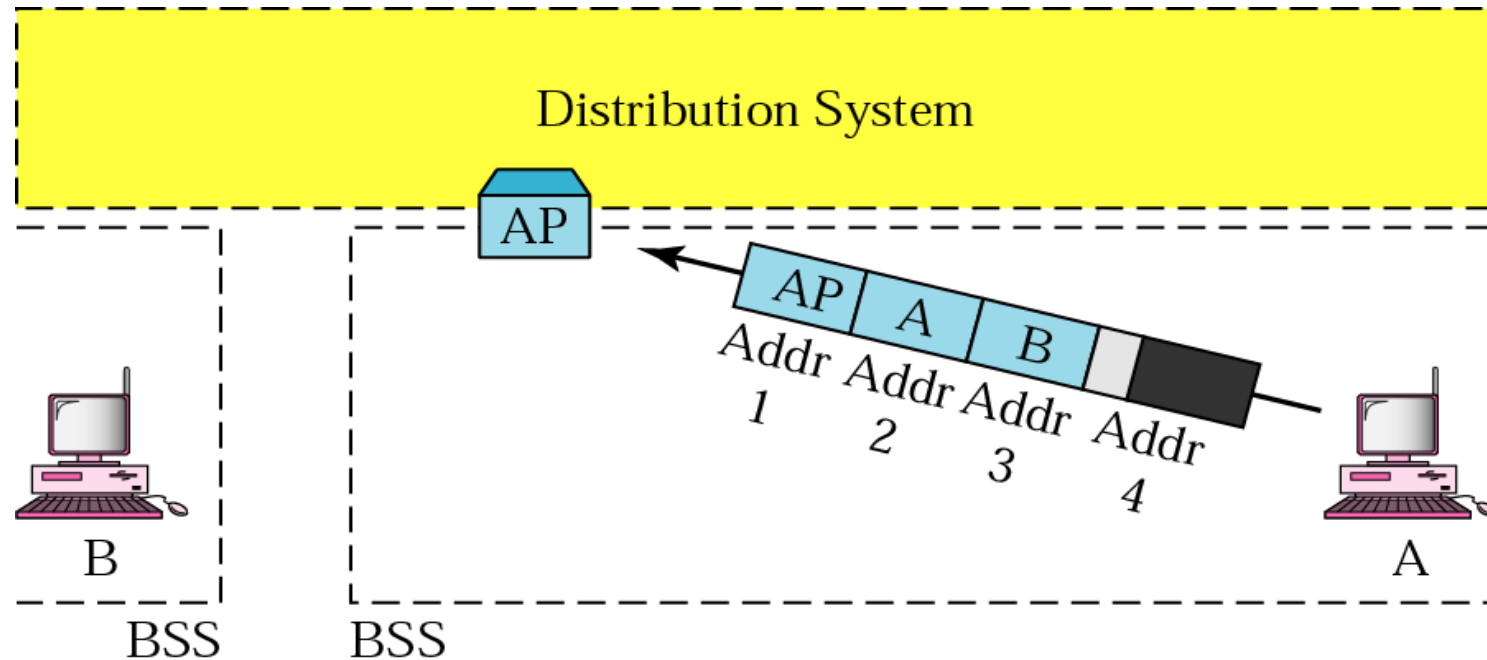
- To DS = 0 and From DS = 0
- Frame is not going to a distribution system
- Frame is going from one station in a BSS to another
- ACK sent to original sender

Addressing mechanism: case 2



- ❑ To DS = 0 and From DS = 1
- ❑ Frame is coming from a distribution system
- ❑ Frame is coming from an AP and going to a station
- ❑ ACK sent to AP

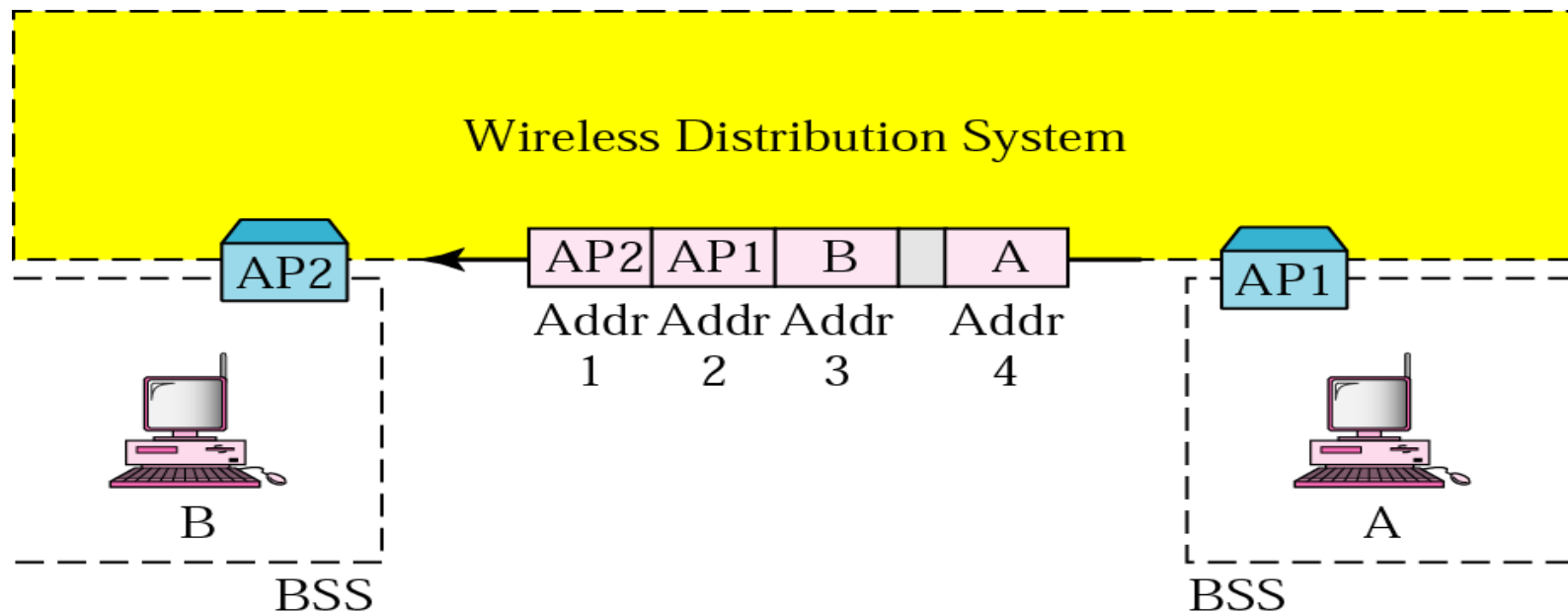
Addressing mechanism: case 3



- ☐ To DS = 1 and From DS = 0
- ☐ Frame is going to a distribution system
- ☐ Frame is coming from a station to an AP
- ☐ ACK sent to original station
- ☐ address 3 contains final destination of frame

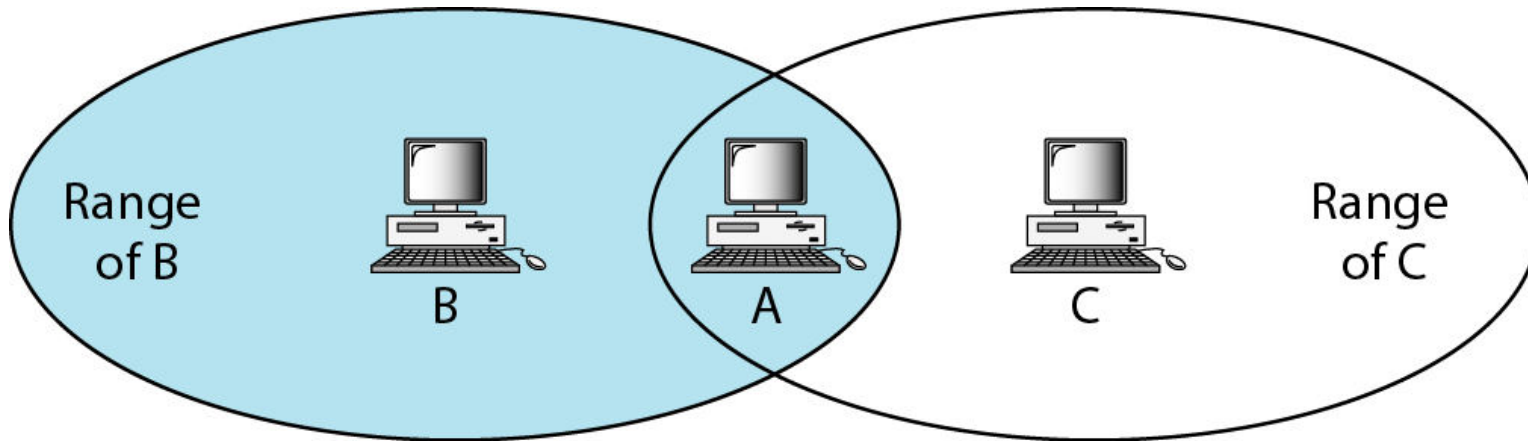


Addressing mechanism: case 4



- ❑ To DS = 1 and From DS = 1
- ❑ Distribution system is wireless
- ❑ Frame is coming from an AP going to another AP
- ❑ ACK sent to original station

Hidden and Exposed Station Problems



B and C are hidden from each other with respect to A.

Figure 14.10 *Hidden station problem*

Hidden and Exposed Station Problems

Note

The CTS frame in CSMA/CA handshake can prevent collision from a hidden station.



Hidden and Exposed Station Problems

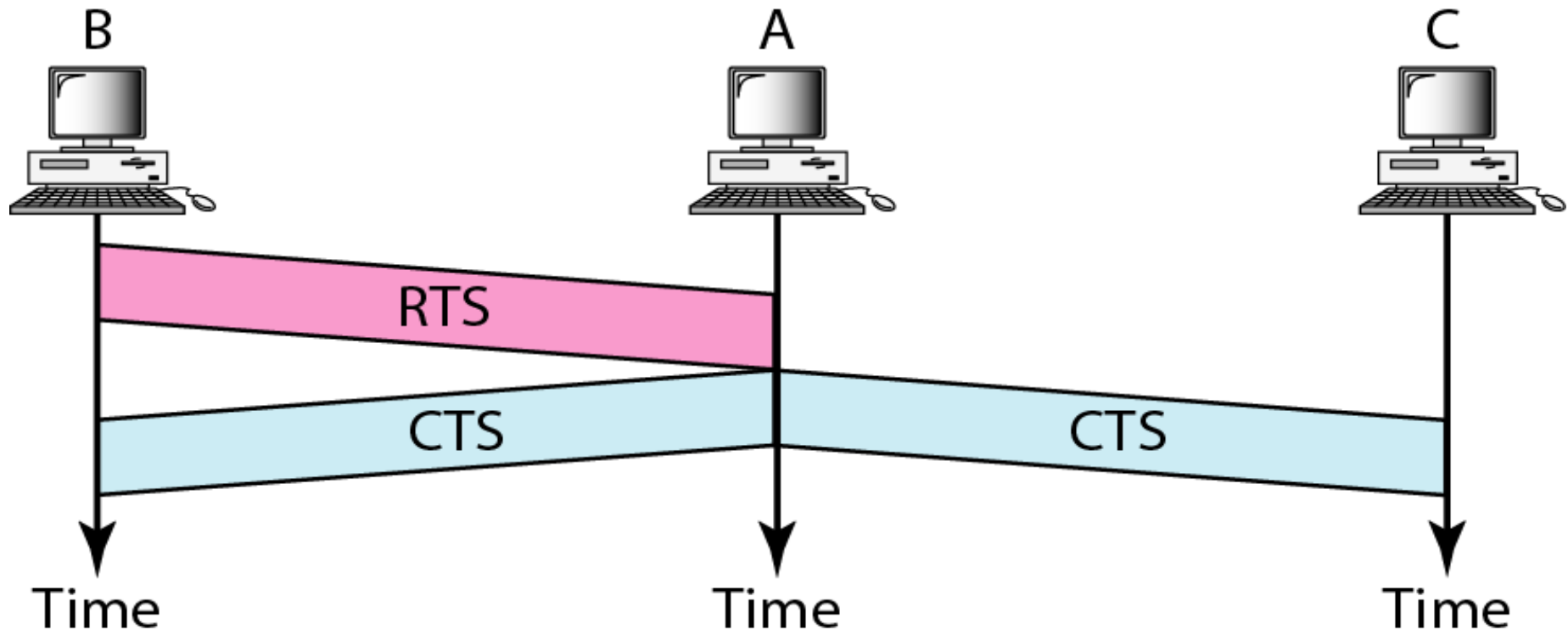
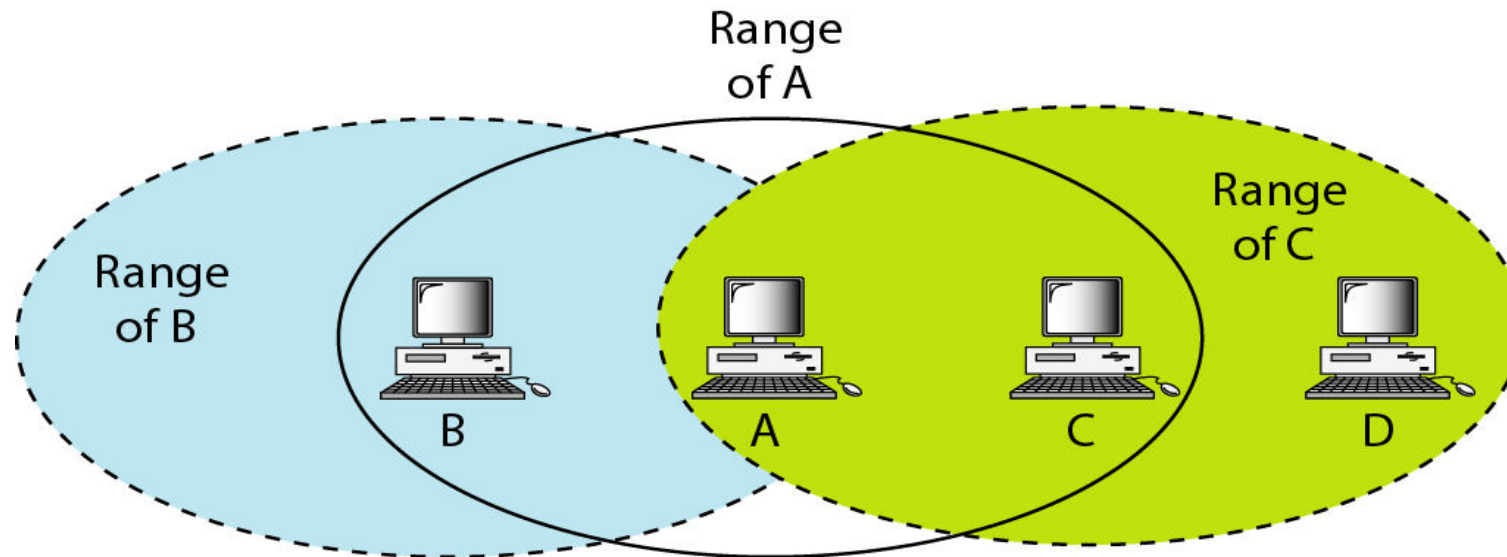


Figure 14.11 *Use of handshaking to prevent hidden station problem*

Hidden and Exposed Station Problems



C is exposed to transmission from A to B.

Figure 14.12 *Exposed station problem*

Hidden and Exposed Station Problems

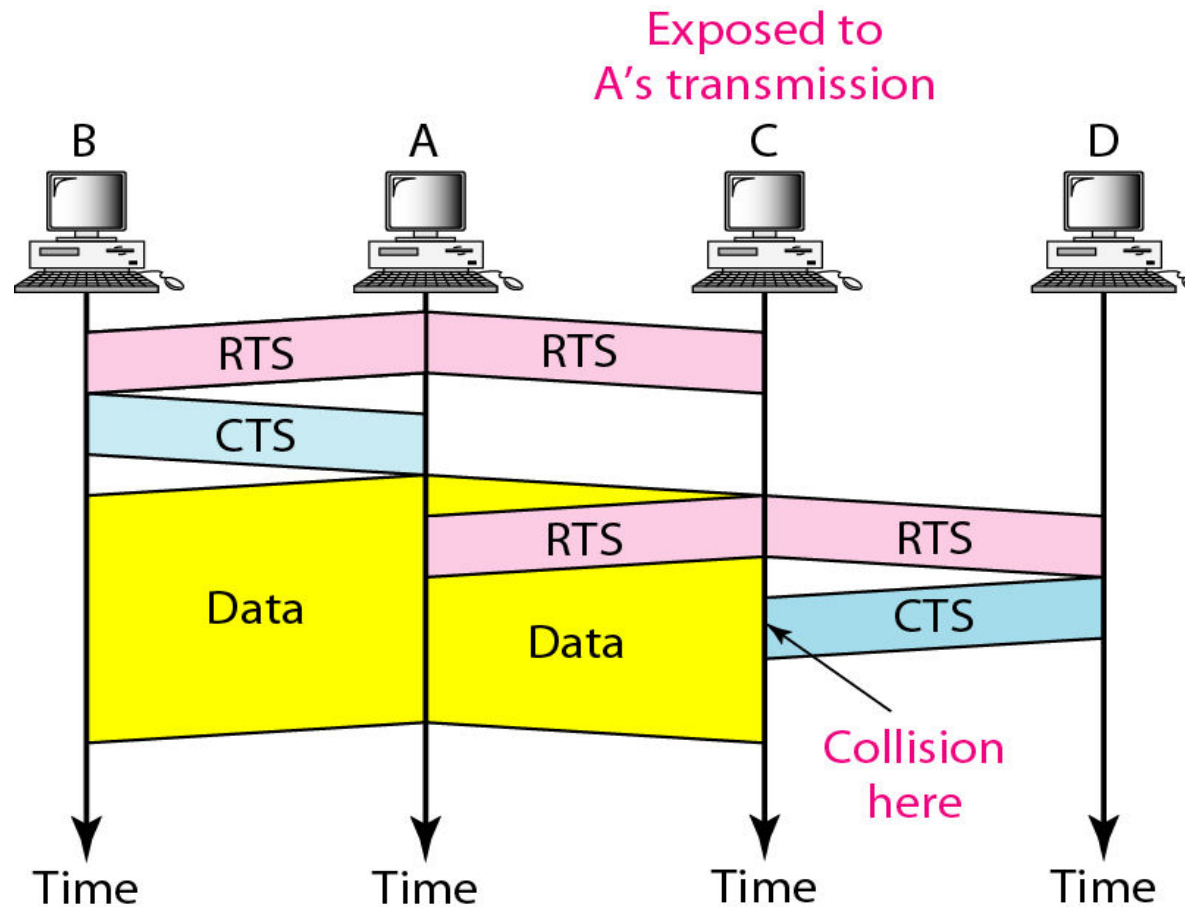


Figure 14.13 *Use of handshaking in exposed station problem*

Physical Layer

Table 14.4 *Physical layers*

<i>IEEE</i>	<i>Technique</i>	<i>Band</i>	<i>Modulation</i>	<i>Rate (Mbps)</i>
802.11	FHSS	2.4 GHz	FSK	1 and 2
	DSSS	2.4 GHz	PSK	1 and 2
		Infrared	PPM	1 and 2
802.11a	OFDM	5.725 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.4 GHz	PSK	5.5 and 11
802.11g	OFDM	2.4 GHz	Different	22 and 54



Physical Layer

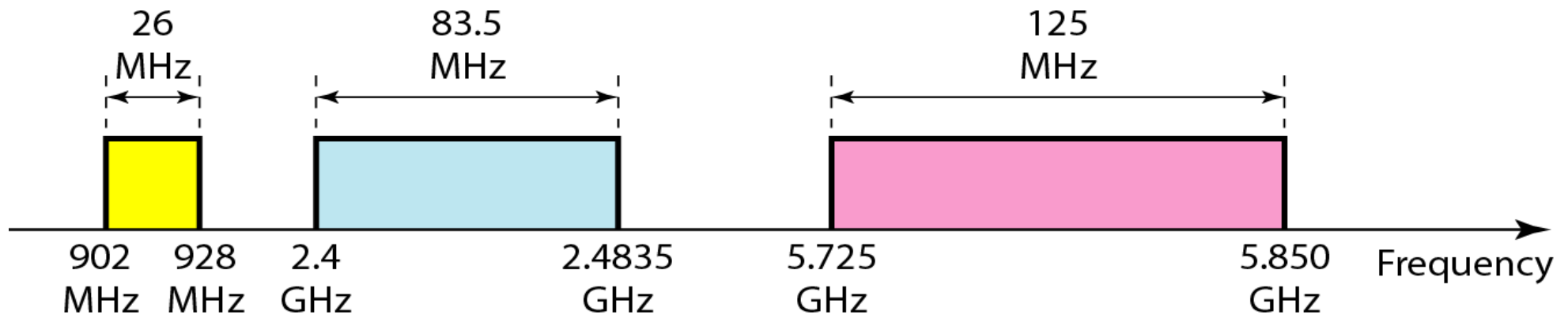


Figure 14.14 *Industrial, scientific, and medical (ISM) band*



Summary

- ❑ The IEEE 802.11 standard for wireless LANs defined two services: basic service set (BSS) and extended service set (ESS).
- ❑ The access method used in the distributed coordination function (DCF) MAC sublayer is CSMA/CA.
- ❑ The access method used in the point coordination function (PCF) MAC sublayer is polling.
- ❑ The network allocation vector (NAV) is a timer used for collision avoidance
- ❑ The Mac layer frame has nine fields. The addressing mechanism can include up to four addresses.
- ❑ Wireless LANs use management frames, control frames, and data frames.
- ❑ IEEE 802.11 defines several physical layers, with different data rates and modulating techniques.



Questions !

